

Book 4 – Annex

Clearing calendar: 1W, 4W and 13W cycles

PRE-AUDIT EDITION PAPER

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Prior Osnias documentation:

Book 1	Architecture	https://www.osnias-clearing.com/architecture/ https://www.osnias-clearing.com/architecture/Osnias_Clearing_Functional_Architecture.pdf
Book 2	Security	https://www.osnias-clearing.com/security/ https://www.osnias-clearing.com/security/Osnias_End_User_Security_Audit_Readiness.pdf
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Book 4	Clearing Rules & Regulatory and Technical Review	https://www.osnias-clearing.com/rules/ https://www.osnias-clearing.com/rules/Livre_4-Osnias_Clearing_Release_1_1_EN.pdf

OSNIAS CLEARING

Book 4 – Annex A

Annex to Clearing Rules – Regulatory and Technical Review, Release 1.1 — September 2026 — Denis Bouzon

Abstract

This annex to Osnias Clearing Book 4 sets the clearing calendar for the three cycle profiles (1W, 4W and 13W). The calendar is counted exclusively in UTC weeks, without reference to months or the civil year, from Monday 21 September 2026, the September equinox. Each cycle ends on a Monday with a 24-hour clearing window, followed by a BURN window of one day in 1W and two days in 4W and 13W, allowing a catch-up day on longer cycles. The year has 52 weeks, 53 every seventh year and 54 every forty-ninth year: a seven-year great year thus has 365 weeks, and the four 13W cycles remain aligned with the solstices and equinoxes. The annex is accompanied by a workbook that applies these rules to any year.

Keywords:

Osnias Clearing; clearing calendar; 1W, 4W, 13W cycles; clearing window; BURN; UTC week; 52-week year; leap week; seven-year great year; 49-year cycle; solstices and equinoxes; multilateral netting

Résumé

Cette annexe au Livre 4 d'Osnias Clearing fixe le calendrier de clearing des trois profils de cycle (1W, 4W et 13W). Le calendrier est compté exclusivement en semaines UTC, sans référence aux mois ni à l'année civile, à partir du lundi 21 septembre 2026, équinoxe de septembre. Chaque cycle s'achève le lundi par une fenêtre de clearing de 24 heures, suivie d'une fenêtre de BURN d'un jour en 1W et de deux jours en 4W et 13W, afin de permettre un rattrapage sur les cycles longs. L'année compte 52 semaines, 53 chaque septième année et 54 la quarante-neuvième année : une grande année de sept ans compte ainsi 365 semaines, et les quatre cycles 13W restent alignés sur les solstices et les équinoxes. L'annexe est accompagnée d'un tableur qui applique ces règles à toute année.

Mots-clés :

Osnias Clearing ; calendrier de clearing ; cycles 1W, 4W, 13W ; fenêtre de clearing ; BURN ; semaine UTC ; année de 52 semaines ; semaine intercalaire ; grande année de 7 ans ; cycle de 49 ans ; solstices et équinoxes ; compensation multilatérale

A.1 Purpose

This annex sets out the Osnias Clearing calendar for the three cycle profiles (1W, 4W and 13W). It complements Book 4, which describes the 24-hour clearing window, finality states and settlement. It is accompanied by a calculation workbook that applies the rules below to any year.

A.2 Principles

- The calendar is counted exclusively in weeks, from Monday 00:00 to Sunday 24:00 UTC. Months and the civil year are not used; the Monday date is the reference date of each clearing.
- Day 1 of the calendar (epoch): Monday 21 September 2026, September equinox, first year of the 49-year cycle.

- The three profiles start together in week 1. Since $52 = 4 \times 13$, the 4W and 13W profiles clear together once a year, in week 1.

A.3 Weekly structure

Day (UTC)	Clearing week	Other weeks of the cycle
Monday	Clearing: 24-hour window (P2P closed; snapshot, statements, disputes 03:00–15:00, settlement from 21:00)	P2P / MINT
Tuesday	BURN (1W, 4W, 13W)	P2P / MINT
Wednesday	BURN (4W and 13W); P2P / MINT (1W)	P2P / MINT
Thursday to Sunday	P2P / MINT	P2P / MINT

BURN duration: one day in 1W, since a member who misses it simply exits the following week; two days in 4W and 13W, so that a member who misses the first day can catch up the next day.

A.4 Cycle profiles

Profile	Length	Clearings	Clearing weeks
1W	1 week	52 per year	every Monday
4W	4 weeks	13 per year	weeks 1, 5, 9, ... 49
13W	13 weeks	4 per year	weeks 1, 14, 27, 40

A.5 Year length and solar alignment

A 52-week year has 364 days, about 1.24 days shorter than the solar year. To keep the four 13W cycles aligned with the solstices and equinoxes, weeks are added as follows:

- 52 weeks in an ordinary year;
- 53 weeks every seventh year, so that a seven-year great year has $6 \times 52 + 53 = 365$ weeks;
- 54 weeks every forty-ninth year, which closes the 49-year cycle.

Additional weeks extend the last cycle of the year: the last 4W cycle lasts 5 weeks (6 in the 49th year) and the last 13W cycle 14 weeks (15 in the 49th year). The 1W profile clears every week, including weeks 53 and 54.

Between two corrections, the alignment oscillates by about one week around the solstices and equinoxes; the residual deviation of the rule is about five days per 49-year cycle.

A.6 13W clearings in year 1

13W clearing	Date (Monday)	Solar marker
Q1	21 September 2026	September equinox
Q2	21 December 2026	December solstice
Q3	22 March 2027	March equinox
Q4	21 June 2027	June solstice

A.7 Start of years (extract)

Year	Rank (7 years)	Length	Start (Monday)
1	1	52 weeks	21 September 2026
2	2	52 weeks	20 September 2027
3	3	52 weeks	18 September 2028
4	4	52 weeks	17 September 2029
5	5	52 weeks	16 September 2030
6	6	52 weeks	15 September 2031
7	7	53 weeks	13 September 2032
8	1	52 weeks	19 September 2033
14	7	53 weeks	12 September 2039
49	7	54 weeks	3 September 2074
50	1	52 weeks	16 September 2075

A.8 Calculation workbook

The attached workbook (Livre-4-Annexe-A-Calendar-cycles.xlsx) applies these rules. Cells on a yellow background are the only inputs: epoch, year displayed and rank of the starting year in the 49-year cycle. For each profile and each day, the grid shows the open activity (Clearing, BURN, P2P / MINT), the Monday date and the week rank within the cycle.